[METRIC]

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SUPERSEDING
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February 1, 1993

COMMERCIAL ITEM DESCRIPTION

BINDER, LOOSE-LEAF, RING: ONE-PIECE COVER

The General Services Administration has authorized the use of this commercial item description for all Federal agencies.

1. Scope. This commercial item description covers loose-leaf binders of one-piece cover construction. The binders are equipped with steel rings for firm retention of papers when closed but which can be readily released by the use of an operating mechanism in the binder. The binders are intended for use in filing records, reports, and other papers when it is desirable that they can be readily removed or inserted.

2. Classification. Binders shall be of the following types as specified:

Type I - Rubber bonded cellulose fiber cover material
Type II - Plastic cover material

3. SALIENT CHARACTERISTICS.

3.1. Design. The binders shall be designed to securely retain filler sheets when rings are closed and shall be such that one or more sheets can be inserted or removed without disturbing the remaining contents. Binders shall be designed so that enclosed sheets will present a flat writing surface for making entries when the binder is open. The binders shall be such that the back and binding mechanism, including the opening lever, or device, at one or both ends of the binding mechanism, will not mar smooth surfaces such as desk, tables, and bookcase shelves.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document, should be sent to: General Services Administration, Federal Supply Service, 26 Federal Plaza, New York N.Y. 10278 ATTN: Engineering Branch (2FYEE).

FSC 7510

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3.2. Construction. The binders shall be built to withstand strains, jarring, and other conditions incident to shipping, storage and normal use. Each binder shall consist of a one-piece cover with a binding mechanism, and the Type I binder may have a binding mechanism and backbone support. Rings shall have positive mating ends when closed through the provision of suitable key depressions or grooves. When mated, ends of rings shall nest smoothly to permit travel of pages back and forth without snagging. The spring or tension plate shall provide sufficient tension to insure rings being mated tightly when closed. Ring metal shall be constructed so that the rings can be opened by means of a lever, trigger, booster, or other device, at one or both ends of the binder.

3.2.1. Binding mechanism. The binding mechanism of the binder shall consist of two ring plates which shall carry the rings, and a spring or tension plate (also referred to as a top or cover plate). The size of the binding mechanism shall be approximately the specified binder sheet size in length and the width shall be controlled by the capacity of the binder in accordance with normal commercial practice.

3.2.2. Ring plates (arch frames). The ring plates, conforming to 3.3.1. herein, shall not be less than 1 mm (0.040") thick for binders having a capacity greater than 2.54 cm (1"); not less than 0.91 mm (0.036") thick for binders having a capacity of 2.54 mm (1") and less, but greater than 6.35 mm (1/4"); and 0.64 mm (0.025") thick for binders having a capacity of 1.27 cm (1/2"). The ring plate shall completely fill the space in the spring or tension plate.

3.2.3. Rings (arches). Rings shall be of steel wire of sufficient hardness to withstand repeated opening and closing associated with heavy usage.

3.2.4. Spring or tension plate (shield). The steel, conforming to 3.3.1. herein, shall be not less than 0.51 mm (0.020") thick for binders having a capacity of 2.54 cm (1") to 3.81 cm (1-1/2"); and not less than 0.38 mm (0.015") thick for binders having a capacity of 2.54 cm (1") to 3.81 cm (1-1/2"); and not less than 0.28 mm (0.011") thick for binders having a capacity of less than 2.54 cm (1"). The spring or tension plate shall be fitted with eyelets through which the fastening rivets will pass. If the spring or tension plate is not used as a cover plate, a cover plate with equivalent characteristic shall be provided.

3.2.5. Cover, Types I and II. The rubber bonded cellulose fiber used in the Type I binder shall be not less than 0.64 mm (0.025") thick for binders up to and including 3.81 cm (1-1/2") capacity, and not less than 0.76 mm (0.030") thick for binders over 3.81 cm (1-1/2") capacity. The plastic used in the Type II Binder shall not be less than 0.89 mm (0.035") thick for binders up to and

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including 3.81 cm (1-1/2") capacity, and not less than 1.40 mm (0.055") thick for binders over 3.81 cm (1-1/2") capacity. Binder covers shall be sufficiently larger than the specified sheet size to protect the filler sheets and index tabs extending up to 1.27 cm (1/2") beyond the outer edge of the filler sheet. To establish a hinge area for the front and back cover, the limits of the backbone area shall be scored by a hot embossing process. The scoring width between the front and back cover shall be not less than the specified capacity of the binder. The Type I binder may, at the option of the manufacturer, have a narrow ridge embossed approximately 6.35 mm (1/4") in from the outside edge, and continuing entirely around all outside edges. Outside corners shall be rounded with a radius of not less than 9.53 mm (3/8").

3.2.6. Binding mechanism and backbone support, Type I binder. When specified in the contract, a strip of pressboard, fiberboard, or rubber bonded cellulose fiber, not less than 0.64 mm (0.025") thick and the length and width of the binding mechanism, shall be placed between the backbone and the binding mechanism. When specified (see 8.6), a support strip shall be extended on each side sufficiently beyond the outside rings and scored for ring protectors to prevent rings from wearing through the covers.

3.2.7. Assembly of binding mechanism. The spring or tension plate containing the ring plates, and the support strip if applicable, shall be attached to the backbone by not less than 2 metal rivets, either one-third tubular or solid, by means of turning over the ends of the rivets onto the front (inside) of the spring or tension plate.

3.2.8. Fly leaves. Two fly leaves shall be supplied with each binder with a capacity of 1.27 cm (1/2") and over. The fly leaves shall be black, fabricated from pressboard, fiberboard or plastic, and shall not extend beyond the covers of the binding side. Fly leaves shall not be less than 0.64 mm (0.025") thick and 4.76 cm (1-7/8") wide for binders 1.27 cm (1/2") to 3.81 cm (1-1/2") capacity, and not less than 1.02 mm (0.040") thick and 7.0 cm (2-3/4") wide for binders over 3.81 cm (1-1/2") capacity.

3.2.9. Exterior marking, lettering, or labeling of covers. The method, location, design, terminology, color, and durability of the marking, lettering, or labeling shall be as specified (see 8.8).

3.2.10. Identification of product marking. The manufacturer's name or trademark, or both (at the option of the manufacturer), shall be printed, stamped, or silk screened, anywhere on the inside of one cover of the binder.

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3.3. Materials.

3.3.1. Steel. The steel shall be of good commercial quality, and all exposed steel surfaces shall have a plated and smooth finish. Ring plates and springs or tension plates shall be made of cold-rolled steel conforming with TABLE I below.

TABLE I

	Thickness	Rockwell Reading	Test Method
Ring Plates (arch frames)	0.64 mm (0.025")	R30T 56.5-67	ASTM E 18
	0.91 mm (0.036")	RB60 minimum	ASTM E 18
	1.02 mm (0.040")	RB60 minimum	ASTM E 18
Spring or Tension Plate (shield)	0.28 mm (0.011")	R15T 80-85	ASTM E 18
	0.38 mm (0.015")	R30T 56.5-72	ASTM E 18
	0.51 mm (0.020")	R30T 56.5-72	ASTM E 18

3.3.2. Type I. Type I binders shall be designed with a cover material made of rubber bonded cellulose fiber conforming to the requirements of Table II below. Rubber bonded cellulose fiber shall be paper base made from cotton fibers, and beater impregnated with SRB rubber. The face side shall have a pyroxylin or vinyl coating, and when specified in contract, material with a pyroxylin face coating shall be coated on the back side with pigment or pyroxylin, as specified (see 8.7). The coating must be suitable for silk screening, stamping, or embossing.

TABLE II. Rubber bonded cellulose fiber requirements

Requirements		Thickness 0.64 (0.025)	Thickness 0.76 (0.030)	Units	Test Method
				mm (inches)	TAPPI T 411
Basis Weight, minimum 61x91cm - 500 sheets (24"x36"- 500 sheets)		513 (315)	570 (350)	g/m^2 (lb_m)	TAPPI T 410
Tensile Breaking Strength, minimum	MD	2800 (160)	3150 (180)	N/100 mm (lb_f/in)	TAPPI T 404
	TD	875 (50)	1050 (60)	N/100 mm (lb_f/in)	

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3.3.3. Type II. Type II binders shall be designed with a cover material made of plastic which must be suitable for silk screening, stamping, or embossing. Unless designated in the contract, plastic cover material shall be antistatic high-density polyethylene, or polyallomer. The polyethylene shall meet the requirements of Type III, Class B, Category 4 of ASTM D 1248. Polyallomer shall conform to the physical properties specified in Table III below.

TABLE III. Physical Properties of Polyallomer

Physical Property	Range of Values	Test Method
Density	0.896 to 0.899 g/cm^3	ASTM D 1505
Tensile Strength, @ yield	21000 to 27000 kPa (3050 to 3850 psi)	ASTM D 638
Bending Modulus	487 to 760 MPa (70000 to 110000 psi)	ASTM D 747
Tensile Impact Strength	230-340 joules (170-250 ft-lb)	ASTM D 1822
Izod Impact Strength @ 23 ^{o}C,	2.3-8.1 joules (1.7-6.0 ft-lb)	ASTM D 256 (method D)
Izod Impact Strength @ 23 ^{o}C	No Break	ASTM D 256 (method E)
Izod Impact Strength @ -10 ^{o}C	9.5-27.1 joules (7-20 ft-lb)	
Brittleness Temperature	-10 to -40 ^{o}C	ASTM D 746
Vicat Softening Temp.	122 to 136 ^{o}C	ASTM D 1525 (Rate B)
Flow rate @ 230 ^{o}C and 2.16 kg load	2.0 - 2.5 g/10 minutes	ASTM D 1238 A or B

3.4. Workmanship. The loose-leaf binders shall be well finished, clean, and all edges shall be smooth. All metal parts shall be free from burrs and rough sharp edges. The binder covers shall be free from warpage, and the hinge scoring depth shall be controlled so that covers will remain closed when the binder is in a flat position on a smooth surface such as a desk or table top. The plastic cover material shall be uniform in thickness, and shall be free from streaks, blisters, scratches, and crazing. The binder shall be free from defects that adversely affect appearance and serviceability or present a hazard to the user.

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4. Quality Assurance. The product provided shall meet the salient characteristics of this document, conform to producer's own drawings, specifications, standards, and quality assurance practices and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5. REGULATORY REQUIREMENTS:

Certification (paperboard). The contractor shall certify that the paperboard used in the manufacture of this item contains a minimum of 20% Post Consumer Recovered Material (PCRM) in accordance with Title 40 of the Code of Federal Regulations Part 247, Comprehensive Guideline for Procurement of Products Containing Recovered Materials; Final Rule. The Government reserves the right to require proof of such conformance prior to first delivery, and thereafter as may be otherwise provided for under the provisions of the contract.

Certification (plastic). The contractor shall certify that the plastic used in the manufacture of Type II binders contains a minimum of 25% Recovered Material (RM) in accordance with Title 40 of the Code of Federal Regulations Part 247, Comprehensive Guideline for Procurement of Products Containing Recovered Materials; Final Rule. The Government reserves the right to require proof of such conformance prior to first delivery, and thereafter as may be otherwise provided for under the provisions of the contract.

6. PACKAGING, PACKING AND MARKING: Packing, packing and marking shall be as specified in the contract or purchase order.

7. REFERENCED DOCUMENTS: The issues of the referenced documents in effect on the date of the solicitation for offers or request for proposals shall be used to determine conformance with the requirements of this Commercial Item Description.

7.1. Title 40, Code of Federal Regulations, Protection of the Environment, Part 247. Copies may be purchased with a VISA, Master Card or prepaid check through: Government Printing Office, Washington, DC; Phone 202-783-3238. When ordering ask for: Title 40, Parts 190-250; Stock Number: 869-013-00-146-0.

7.3. ASTM standard test methods can be obtained from the American Society for Testing Materials; 100 Barr Harbor Dr., West Conshohocken, PA 19428-2959.

7.4. TAPPI standard test methods can be obtained from Technical Association of the Pulp and Paper Industry; Technology Park/Atlanta P.O. Box 105113; Atlanta, GA 30348.

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8. ORDERING DATA. Purchaser shall specify the following in the contract or purchase order:

8.1. Title number and date of this commercial item description.
8.2. Type of binder required.
8.3. Back coating of Type I binder, if required.
8.4. Plastic for the Type II cover. (polyethylene or polyallomer).
8.5. Specify color, if other than black is required; and specify special grain or finish, if required.
8.6. If Type I binders are to have binding mechanism and backbone support is to be extended to provide ring protector strip.
8.7. Whether rubber bonded cellulose fiber with pyroxylin face coating is coated on the back side by pigment or pyroxylin.
8.8. Exterior marking, lettering or labeling of cover(s).
8.9. Packaging, packing, and any special marking for required.

MILITARY INTEREST:

DoD has no registered interest in revisions or amendments to this Commercial Item Description until further notice

PREPARING ACTIVITY:

GSA-FSS

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